INTRODUCTION TO RESUBMISSION

I appreciate the thoughtful review of my K01 application and the reviewers' strong enthusiasm for the scientific significance of the proposed research, high-quality mentorship team, and alignment of my background and career development plan. The reviewers also noted the timeliness of my application considering upcoming updates to the national physical activity (PA) guidelines. Thus, the overarching goal of my revised application remains the same: to advance understanding of the 'PA health paradox' by rigorously investigating the relationship between occupational physical activity (OPA) and cardiovascular disease (CVD) and directly inform the development of evidence-based PA guidelines and workplace policies for this important and understudied exposure group. Below, I describe how I have addressed score driving weaknesses in the research methodology, training plan, and analytic approach in this resubmission application.

Secondary analysis of the proposed cohorts do not directly address the gaps. *Response:* I added a new Aim 1 focusing on primary data collection to directly fill a methodological gap in the current literature and removed less impactful secondary analysis of one cohort study (nuMoM2b-HHS cohort). My new study will validate self-reported OPA measures used in Aim 2 against actual device-based metrics in a sample of working adults and provide a foundation for future research. This calibration study directly addresses research gaps in OPA measurement to improve interpretability of future secondary cohort analyses, enhance the methodological rigor of this proposal, and support a more comprehensive training plan. Additionally, the two cohorts now examined in Aim 2 provide a unique and timely opportunity to rigorously examine the PA health paradox mechanisms with novel, high-quality exposure assessment and biologically relevant outcomes.

Career development plan based on secondary analyses has overlap with previous training. *Response:* The revised Training Goal 1 now includes new, hands-on training with primary data collection in an observational study. Methods will include wearable device deployment, data integration, and validation methodology. This training is also supported by an additional mentor, Dr. Keadle, a PA measurement expert. In addition to the primary data collection, training in cohort coordination has been expanded through active participation in the CARDIA and RURAL cohort infrastructures which are likely established cohorts for future ancillary studies examining OPA. These activities, reflected in Training Goal 1 (section D.1), provide critical new skills not that better align with my long-term goal of leading ancillary studies in large cohort settings.

Complexities of associations based on potential confounders need to be discussed in more detail. *Response:* While the revised Aim 2 retains the secondary analyses in the CARDIA and REGARDS cohorts from the original proposal, I expanded the analysis plan to include improved strategies to evaluate potential effect modification by socio-demographic (i.e., sex, age, and race) and behavioral factors (i.e., leisure-time physical activity [LTPA], smoking status, and alcohol intake). The impact of these covariates will be explored via a series of models with progressive adjustment as well as stratified analyses across exposure categories to examine potential confounding and effect modification. The details are now outlined in the following Aim 2 analytic approach sections: Study 1 - G.2.III and Study 2 - G.2.VI.

Justification for inconsistencies in OPA measurement methods (device vs. self-reported) is lacking. *Response:* The revised proposal includes a calibration study in Aim 1 that enables me to identify the most valid single or combination of self-reported OPA questions and previous day PA recall for estimating device-based OPA. This will be completed prior to Aim 2 to help standardize measurement methodology and exposure classification in the subsequent proposed secondary analyses and future studies. In addition to this calibration step, I have clarified the importance of considering both self-report and device-based OPA exposures. Both measurement modalities are available in existing cohorts and reported in the literature, though there is considerable variability in the measurement of OPA exposure profiles. Understanding whether different measures impacts associations between OPA and health outcomes will help to clarify the mixed evidence on risk relationships between OPA and health and for selecting optimal measurement methods for future studies.

Insufficient power analysis calculations are presented. *Response:* Detailed power calculations are now included for both the Aim 1 validation study and Aim 2 cohort analyses. The selected sample size for Aim 1 (n=50) provides ≥80% power to detect expected correlations between self-reported and device-based OPA. For Aim 2, the CARDIA and REGARDS cohorts have sufficient power to detect clinically meaningful differences in blood pressure and CVD biomarkers, and power to detect modest hazard ratios for major CVD events. The detailed power analyses are described in the following sections: Aim 1 - G.1.III, Aim 2 - G.2.IV and G.2.VII.

Inadequate discussion of start-up funds/support. *Response:* The application now includes a summary of institutional support, including a two-year internal pilot grant, $25,000 start-up research funds, equipment and lab infrastructure for data collection, and early career development programs. These resources, now outlined in departmental and institutional support (section D.4), ensure adequate support to complete the proposed K01.

PROJECT SUMMARY

This K award application seeks to advance our understanding of the 'physical activity health paradox,' which postulates that, contrary to the known health benefits of leisure-time physical activity (LTPA), occupational physical activity (OPA) may not protect against cardiovascular disease (CVD). In some studies, OPA is even associated with significantly increased cardiovascular risk and mortality. If confirmed, there are important implications of this paradox: 1) workplace policies to mitigate the health consequences of OPA may need to expand their scope to include reduction of cardiovascular risks, and 2) physical activity guidelines will need to explicitly acknowledge and communicate that time spent in OPA does not contribute to 'protective' time 'being active' and the related cardiovascular health benefits. The upcoming 2028 revision of the 2018 Department of Health and Human Services (DHHS) Physical Activity Guidelines makes right now a critical window for generating new, rigorous evidence that could inform policy and change how OPA is included in recommendations. However, the development of targeted interventions and guidelines for workers is hampered by poor measurement of OPA that uses highly variable, unvalidated self-report instruments or broad categorizations of job characteristics. Further, a dearth of rigorous empirical research investigates outcomes pursuant to understanding underlying biological mechanisms of OPA as it relates to CVD. This proposed career development project will address these gaps through two aims: (1) conduct a primary primary data collection observational study to validate self-report and previous day recall OPA measures against best practice device-based metrics to identify the best self-report OPA measure(s) to use in future large-scale research, and (2) test associations between OPA and sub-clinical and clinical CVD outcomes using two large, well-characterized NHLBI-funded population-based cohort studies (CARDIA and REGARDS). The project outputs will include identification of the best self-report OPA measures for use in future studies and at least four high-impact, peer-reviewed articles. Findings will inform the standardization of OPA assessment tools and clarify potential cardiovascular harms associated with high OPA. The proposed training plan will prepare me to lead ancillary studies within large cohorts and includes mentorship from an experienced team at West Virginia University, the West Virginia Medicine Heart and Vascular Institute, the University of Alabama at Birmingham, and California Polytechnic State University. The training plan will develop my skills in 1) conducting primary data collection of OPA exposures using the spectrum of available measurement methods, 2) leading ancillary studies in multi-center national cohorts, 3) collecting and interpreting sub-clinical and clinical CVD outcomes, and 4) professional grant writing and project management. These training goals and research aims will lay the groundwork for an independent, extramurally funded research career committed to elucidating the disproportionate burden of CVD among workers with high OPA.

PROJECT NARRATIVE

My project seeks to better understand why it appears that physical activity occurring at work (i.e., manual labor) does not provide the same heart health benefits as leisure time exercise and may even be harmful. I will first identify the best way to accurately measure activity at work and then use large health studies to examine how these measures of occupational activity relate to heart health. This research could help improve workplace policies and future physical activity guidelines to better protect the cardiovascular health of workers with active jobs.

A) CANDIDATE BACKGROUND

As an early career faculty member at West Virginia University (WVU), I am committed to becoming a leader in advancing cardiovascular health among individuals in physically demanding occupations, an understudied and high-risk population. With specialized training in exercise science, epidemiology, and three years of postdoctoral training at Centers for Disease Control and Prevention's National Institute for Occupational Safety and Health (CDC/NIOSH), I am uniquely positioned to build a research program at the intersection of occupational health, physical activity (PA) epidemiology, and cardiovascular disease (CVD). I have established myself as a productive researcher with 44 peer-reviewed articles (16 first-author) to date. This grant will provide me with the needed training in primary data collection, cohort study management, clinical cardiovascular measures, and grant writing as well as the protected time and mentorship necessary to establish an independent research portfolio, with the goal to positively impact worker health.

A.1) Graduate Training:

I completed my graduate training at the University of Pittsburgh where I earned a PhD in Exercise Physiology with a concentration area in Epidemiology and Biostatistics. I worked primarily as a research assistant mentored by Dr. Bethany Barone Gibbs on an NIH/NHLBI-funded randomized controlled trial testing the impacts of a workplace sedentary behavior reduction intervention on blood pressure (RESET BP, R01 HL134809). I trained directly in device-based PA measurement, acute cardiovascular responses to PA , CVD risk factors, and workplace intervention delivery. My dissertation project combined this experiential learning with my interest in occupational health to examine the biological mechanisms of the 'PA Health Paradox', the seemingly contradictory evidence that occupational PA (OPA) may not have the same health-enhancing effects as leisure-time PA (LTPA). My dissertation examined acute cardiovascular responses to OPA in workers with physically demanding jobs. We found that workers were meeting or exceeding US PA Guidelines for aerobic PA during work alone.[1] However, their 24-hour blood pressure and heart rate were elevated on workdays compared to non-workdays without OPA.[2] These cardiovascular responses were also exaggerated by low worker fitness and high work-related stress.[3] Because chronic exposure to increased cardiovascular loads is expected to lead to cardiovascular damage, these findings provided the first empirical evidence for the proposed biological mechanisms to explain the paradoxically opposing health impacts of OPA and LTPA.

A.2) Post-doctoral Training:

I completed my post-doctoral training at the CDC/NIOSH from July 2019 to August 2022. During this time, I successfully led several small internal CDC grant applications and projects examining cardiorespiratory responses to OPA in a variety of work settings such as healthcare and public safety.[4–9] My work on the CDC COVID-19 response and related first-author report was recognized by receipt of the 2022 US Food and Drug Administration Honor Award and the 2023 US Public Health Service Citation Award.[10] Most importantly, this training taught me the value and challenges of conducting impactful occupational health research.

During this time, I continued examining the cardiovascular impacts of OPA using population-based data. My cross-sectional analysis of the National Health Interview Survey demonstrated that self-reported OPA was directly associated with coronary heart disease in the US workforce.[11] In an analysis of the Coronary Artery Risk Development In Young Adults (CARDIA) Cohort, I published descriptive data on device-based PA patterns across occupational groups, identifying occupation as a strong determinant of individual PA.[12]

In addition to my research, I also served as Co-Assistant Coordinator of the NIOSH National Occupational Research Agenda (NORA) Public Safety Sector Program and Council. In this role, I interacted with scientists, stakeholders, and workplace health practitioners to organize and host regular meetings to facilitate sharing and dissemination of information and best practices for workplace health. This experience was invaluable to learning the importance of collaboration in developing and disseminating occupational health research.

A.3) Early Career Faculty Achievements:

Since joining WVU's School of Public Health in August 2022, I continued my professional development and research into the health impacts of OPA. I successfully competed for an internal Research Scholars Career Development grant sponsored by the West Virginia Clinical and Translational Science Institute (WVCTSI). The two-year award supports the development of K applications by providing 50% protected research time, mentorship, professional development, and resources to conduct pilot research. During the program, I published three first-authored, peer-reviewed articles,[13–15] conducted pilot research exploring novel, device-based OPA measurement methodology, attended professional development events, and completed a K grant writing program to develop this application. While the program has provided me with a strong foundation to take full advantage of the opportunities offered by the K01 but also enabled identification of my gaps and made me realize my need for the mentored training that I have outlined in this application.

B) CANDIDATE CAREER AND TRAINING GOALS

My ultimate career goal is to be a leader in the field with extramurally-funded research studying OPA and cardiovascular health in large cohorts. This award will provide me the protected time and funding to access training and travel necessary for goals which would not be otherwise possible. Although I have some experience and initial publications in my field, I require additional training to reach my goals. Notable training gaps include 1) Large-scale study design/administration: while I've conducted analyses using multi-center cohort data, I have relied on existing data with no experience developing and administering large studies, especially with remote data collection or in ancillary studies; 2) Advanced training and understanding of CVD outcomes: as an exercise physiologist, I was trained in upstream physiological mechanisms and preventative medicine. I have a training gap in measuring and interpreting clinical and sub-clinical CVD outcomes that are imperative for advancing my research agenda; and 3) Formal training in grant writing and leadership: although I have been successful in obtaining and managing small internal grants from my home university and at CDC/NIOSH, I must advance my skills in professional grant writing to successfully compete for federal R-level grants. Through my proposed mentor support and planned research outputs, I will be able to gain experience conducting impactful research and develop rigorous and impactful future grant applications proposing purpose-built ancillary studies within established cohorts. Reflecting these needs, my training goals are:

Training Goal 1 - Gain experience conducting and managing research with remote data collection and within large, federally funded, multi-center cohort studies

Training Goal 2 - Obtain understanding and experience in measuring and interpreting clinical and sub-clinical cardiovascular disease outcomes

Training Goal 3 – Develop professional skills in grant writing, research management, and leadership

C) MENTORSHIP PLAN

C.1) Mentorship and Training Activities: My goals will be achieved in collaboration with my mentors who have a plethora of content-specific expertise to enhance the project (ambulatory PA measurement, occupational epidemiology, cardiovascular epidemiology, PA epidemiology, and clinical cardiovascular imaging). To facilitate training from the mentorship team, I will meet with my primary mentor (Dr. Smith) weekly, individual members at least monthly, and the entire team at least quarterly to discuss ongoing training and project tasks/goals. My mentors, the training goals and research aims associated with each, meeting frequency, and their related expertise are outlined in Table 1. In addition to my mentorship team, WVU is an ideal setting to support my training and research as a land grant, R1 institution with close connections to the WVU Heart and Vascular Institute in a state with disproportionately high rates of CVD.

C.2) Gordon Smith, MD, PhD (primary mentor): Dr. Smith, a Distinguished Endowed Professor at WVU, is an esteemed physician-epidemiologist with a focus on occupational health and injury research and over 240 publications. His expertise spans large database analysis, surveillance system development, and primary data collection. Along with his extensive track record of federal research funding from the NIH, CDC, and the Department of Defense, he has a stellar record of both formal and informal mentorship, including guiding many junior faculty in successful K Awards, large grant applications, and independent research careers. As the primary mentor, Dr. Smith will provide strategic oversight, advice, and training to enhance research execution, grant writing, and leadership. His primary role will be to coordinate the multidisciplinary mentorship team and to direct and support my overall professional development toward research independence and extramural funding.

C.3) Bethany Barone Gibbs, PhD, FAHA: Dr. Gibbs, a professor and my department chair at WVU, is an expert in CVD epidemiology and lifestyle interventions for cardiometabolic disease prevention. With over 190 publications and significant funding as PI

Table 1 – Career Development Mentors for Dr. Tyler Quinn

Name/Institution	Role	Associated Aims/Goals	Meeting Frequency	Relevant Expertise
Gordon Smith, MD MPH *West Virginia University, School of Public Health*	Primary Mentor	Aim: 1 & 2 Goal: 1-3	Weekly	• Extensive leadership, grant writing, & project management experience
Bethany Barone Gibbs, PhD *West Virginia University, School of Public Health*	Co-Mentor	Aim: 1 & 2 Goals: 1 & 3	Monthly	• Physical activity & cardiovascular epidemiology
Kelley Pettee Gabriel, PhD *U of Alabama at Birmingham, School of Public Health*	Co-Mentor	Aim: 1 & 2 Goal: 1 & 2	Monthly	• CARDIA & REGARDS ancillary PI • Ancillary study management
Sarah Keadle, PhD MPH *Cal Poly State University, Kinesiology and Public Health*	Co-Mentor	Aim: 1 Goal: 1	Monthly	• PA measurement development • Wearable sensors
George Sokos, DO *West Virginia University Heart and Vascular Institute*	Co-Mentor	Goal: 2	Monthly	• Cardiovascular imaging • Clinical HF & CVD outcomes

from NHLBI and AHA, she has demonstrable expertise in executing large-scale studies deploying device-based and self-reported PA assessments as well as assessing CVD outcomes, essential for this proposal. She is a committed mentor, having guided many junior faculty, PhD and master's candidates, and a postdoc to research success. She will provide mentorship for Aims 1 & 2 execution and training goals 1 & 3, focusing on the design and execution of high-quality human subject research.

C.4) Kelley Pettee Gabriel, PhD, FACSM, FAHA: Dr. Gabriel is a professor in the Department of Epidemiology and Associate Dean for Research at the University of Alabama at Birmingham (UAB) School of Public Health with over 250 peer-reviewed publications. She trained in exercise physiology and epidemiology with expertise implementing PA measures in large, observational studies. She has a strong mentorship history and significant funding from NHLBI, NIA, and AHA. Dr. Gabriel is MPI of the CARDIA Birmingham Field Center (2023-33) and PI/MPI of ancillary studies in several large, federal cohorts including, REGARDS CARDIA, Jackson Heart Study (x2), Multi-ethnic Study of Atherosclerosis (MESA), and leads the PA and sedentary behavior assessment in other studies including the Study of Women's Health Across the Nation (SWAN). Dr. Gabriel's experience will provide topic expertise in Aim 1, direct connections with the CARDIA and REGARDS coordinating centers for the Aim 2 analyses, and expert mentorship and connections for training goals 1 & 2.

C.5) Sarah Keadle, PhD, MPH, FACSM: Dr. Keadle is an associate professor of Kinesiology at California Polytechnic State University and a leading expert in PA and sedentary behavior measurement. Her research focuses on improving methods for assessing movement behaviors using wearable sensors and validated self-report instruments. She played a key role in developing and validating ACT24, a previous day PA recall tool I will utilize in Aim 1, and has led many studies evaluating PA and sedentary time using accelerometry in free-living settings.[16,17] As a co-mentor, she will provide mentorship and methodological guidance on the design and execution of the Aim 1 validation study and training goal 1. Her expertise in PA measurement and data integration will be pivotal in identifying the optimal self-report OPA measure(s) for future large-scale studies.

C.6) George Sokos, DO, FACC: Dr. Sokos, a cardiologist and director of the advanced heart failure program at the WVU Heart and Vascular Institute, will serve as a co-mentor focusing on training goal 2. With his extensive clinical expertise in CVD, advanced heart failure, and transplant cardiology, as well as his research focus on abnormalities of heart failure, cardiomyopathy, and heart failure risk assessment, Dr. Sokos is well-positioned to guide me in this area. His connections to the field of clinical cardiology, diverse educational background, including degrees in pharmacy and medicine, combined with his experience as a program director and professor, further strengthens his ability to provide valuable mentorship throughout the project.

C.7) Scientific Advisory Board: In addition to the formal mentorship team, a scientific advisory board will be formed to join a meeting with the entire mentorship team at least twice each year to discuss project progress, plans, and challenges. The board will include Virginia Howard, a distinguished professor and neuroepidemiologist at UAB with a long-standing career as a PI of many US cohort studies including REGARDS. The board will also include Els Clays, associate professor at Ghent University and a leading expert in studying the PA health paradox in European cohorts.

D) TRAINING PLAN

D.1) Training Goal 1 - Gain experience conducting and managing research with remote data collection and within large, federally funded, multi-center cohort studies (Drs. Smith, Gabriel, Gibbs, & Keadle)

My career goal is to conduct large-scale studies within established cohort studies, investigating the cardiovascular health impacts of OPA. While I have assisted with primary data collection within randomized controlled trials and small cross-sectional studies, managing large-scale data collection within a multi-center cohort study requires specialized skills, knowledge, and expertise. To demonstrate my ability and feasibility of methodologies, I will conduct primary data collection with remote deployment of best-practice device-based and self-reported OPA assessments, conduct two approved ancillary studies in large cohorts (secondary analyses, see letters of support), and engage in investigator training activities within large cohorts. This goal will be directly mentored by Drs. Smith, Gabriel, Gibbs, and Keadle, all of whom have successfully led cohort studies as PI. Dr. Smith will provide individualized mentorship and coordinate the training across the multidisciplinary mentorship team while the co-mentors will provide direct mentorship related to the specific studies for which they are involved. This training/mentorship experience will include the following activities:

I. Mentorship meetings with Dr. Smith: To facilitate my training in this goal, I will hold weekly meetings with Dr. Smith and monthly mentoring meetings with Drs. Gabriel, Gibbs, and Keadle where I will specifically discuss issues and progress related to my training in research methods and procedures.

II. Participation in established cohort operations committees: To gain hands-on training in large-scale

cohort operations, I will participate in at least two operations committees in ongoing federal cohorts during the award. Final committee selection will be based on need and fit. Dr. Gabriel, chair of the overall CARDIA Operations Committee which includes the Ancillary Studies, Participant Retention, and Exam subcommittees among others, will facilitate connections via her leadership in CARDIA, REGARDS, and other national studies.

III. Conduct primary remote data collection: As part of Aim 1, I will gain hands-on experience in the design and implementation of primary data collection in an observational study using best-practice OPA measurement methods, including remote deployment of wearable monitors (activPAL) and integration with self-report instruments and time-use diaries. This process will provide training in all aspects of data collection logistics, including recruitment of working adults, mail-based monitor delivery and retrieval, participant instruction and compliance monitoring, secure data transfer and storage, and quality control procedures for accelerometry data. I will work closely with my mentor, Drs. Gibbs, who has led several similar remote monitoring studies, to develop participant materials and protocols that can be scaled to future ancillary studies within federal cohorts.

IV. Methods and applications of cohort studies course: I will build my knowledge of cohort study design, implementation, and analyses by completing the synchronous online professional course, *Methods and Applications of Cohort Studies* (340.706.79) offered by the Johns Hopkins Bloomberg School of Public Health.

V. Observe cohort study data collection/management: I will expand my understanding of cohort study management by directly observing study operations (e.g., data collection, staff management). I will visit UAB for one week with PIs Drs. Kelley Gabriel, Virginia Howard, and Suzanne Judd, who are involved with multiple national cohorts (see support letters). Depending on funding and scheduling, I will observe studies such as CARDIA, RURAL, REGARDS, and Jackson Heart, all of which are expected to be active during this award.

D.2) <u>Training Goal 2 - Obtain understanding and experience in measuring and interpreting clinical and sub-clinical cardiovascular disease outcomes (Drs. Sokos and Gabriel)</u>

I will engage in didactic and applied training in the measurement and interpretation of non-invasive measures of CVD (e.g., echocardiogram and cardiac computed tomography [CT]) as well as in measuring and adjudicating CVD outcomes in large studies. This training will support my ability to use these outcomes effectively in future research. I will achieve training goal 2 via the following activities:

I. Mentorship meetings with Dr. Sokos: I will meet with Dr. Sokos, the director of the advanced heart failure program at the WVU Heart and Vascular Institute, at least monthly to discuss and organize training progress, ongoing analyses, and catalyze learning. These meetings will be held throughout the performance period.

II. Cardiopulmonary physiology and cardiology courses: In preparation for specialized cardiovascular imaging and CVD outcomes training, I will complete two courses in the WVU School of Medicine: Advanced Anatomy, Physiology, and Pathophysiology (NSG 744A) and Cardiology for Nursing (NSG 435). These courses will provide advanced instruction of important concepts involved in cardiopulmonary physiology, treatment, and pathophysiology, allowing me to more fully understand concepts covered in subsequent specialized training.

III. Echocardiography technician training: Following the coursework, I will attend selected sessions of the 12-month echocardiography technician certificate program offered at WVU Hospital. While I will not need to become a certified technician, the sessions which support understanding of measurement methods, image interpretation, and clinical meaningfulness will be valuable to develop my understanding as a researcher. Dr. Sokos and I will determine which sessions will be most appropriate for me to attend throughout my training.

IV. Individualized cardiac imaging training and observation: With a fundamental understanding built, I will then attend one-on-one sessions with Dr. Sokos and/or the director of cardiac imaging at WVU, Dr. Lakshmi Muthukumar. We will discuss the foundational concepts of cardiac imaging, cardiovascular anatomy and physiology, as well as observe and practice reading clinical echocardiogram and CT images.

V. CARDIA outcomes surveillance and adjudication subcommittee: Throughout the first four years of the award, I will get direct experience in methods used to adjudicate clinical CVD outcomes in a large multi-center cohort study by participating in the CARDIA Outcomes Surveillance and Adjudication Subcommittee (OSAS). OSAS, consisting of clinicians from all CARDIA centers, oversees the morbidity and mortality definitions, collection, and adjudication processes for all clinical cardiovascular and cerebrovascular outcomes. Under the direction of Dr. Gabriel, I will participate in OSAS meetings to gain a practical understanding of the endpoint collection process so that I may utilize such measures more effectively in future research.

VI. 10-day seminar on epidemiology & prevention of CVD & stroke: This is an educational program sponsored by the American Heart Association (AHA) aimed at enhancing participants' careers in CVD and stroke research through skill development in grant writing, manuscript development, teaching, and clinical practice. The program features interactive learning with expert epidemiologists and prevention researchers.

D.3) Training Goal 3 - Develop grant writing, research management, and leadership skills (Dr. Smith)

My ultimate career goal is to establish a robust, externally-funded research program focused on OPA and cardiovascular health. An ancillary study adding enhanced OPA measurement within a large established NIH-funded cohort is an efficient, impactful way to examine the PA Health Paradox. To develop a competitive R01-level funding application and conduct such a study, I will need to improve my grant writing, develop my project planning skills, and conduct pilot data collection. I will develop my skills in the following ways:

I. Regular mentorship meetings and conduct grant reviews: In addition to weekly meetings with Dr. Smith, I will reserve quarterly meetings with him, increasing to monthly or bi-weekly by the third year of this award, to discuss grant writing, project planning, and related topics like data generation and partnership development. I will also participate in NIH grant review as an early investigator (I was accepted into program in April 2025). The goal is to submit an R-level grant applications to NIH (likely NHLBI) by the end of year 4.

II. WVCTSI PI Academy: I will participate in the WVCTSI PI Academy, a monthly seminar series with multidisciplinary experts focusing on research project development, management, and execution.

III. R01 grant writing group: WVCTSI's director of professional development facilitates a 15-week grant writing group each year. The course includes six, 2-hour sessions to provide guidance on the required sections of the application, peer-to-peer feedback, and editorial mentorship. It culminates in a competitive R01 submission. I will participate in year 3 to enable my first R01 submission in year 4 with time for resubmission.

IV. Professional conferences: I will attend at least two conferences each year such as the AHA Epidemiology|Lifestyle Conference, American College of Sports Medicine (ACSM) Annual Meeting, International Society of Behavioral Nutrition and Physical Activity (ISBNPA) Annual Conference, or International Commission for Occupational Health (ICOH) Bi-Annual Congress (Table 2). These meetings will provide opportunities to disseminate research findings, foster collaborations, and meet thought leaders.

V. Workgroups on occupational health, physical activity, and CVD: I will attend and present research to the ICOH CVD sub-committee and CARDIA Physical Activity Working Groups. These committees engage in community/stakeholder outreach and share ongoing research.

D.4) Departmental and institutional support: My department has committed 75% protected time for research if the award is funded, with a reduced teaching load (two courses/year). I also received $25,000 in start-up funds and a two-year internal pilot grant with data collection to prepare for this current application. These resources, along with access to early career development programs, ensure strong institutional support for the proposed K01 training and research. Lastly, co-mentor Dr. Gibbs has a full functioning accelerometry laboratory with activPAL devices and software which will be available to support Aim 1 data collection.

Table 2 - Training Timeline for Dr. Tyler Quinn

Training Goal 1 - Gain experience conducting and managing research with remote data collection and within large, federally funded, multi-center cohort studies	**Y1**		**Y2**		**Y3**		**Y4**		**Y5**	
Weekly meetings with Dr. Smith and monthly meetings with co-mentors										
Participation in operations committees within established cohorts (Dr. Gabriel)										
Design/conduct primary data collection with best-practice remote OPA monitoring with (Dr. Gibbs)										
Methods and applications of cohort studies course at Johns Hopkins University (340.706.79)										
Site visit and observation of ongoing cohort study activities at UAB (Dr. Gabriel)										
Responsible conduct of research training (RCR) (sponsored by WVCTSI)										
Training Goal 2 - Obtain understanding and experience in measuring and interpreting sub-clinical cardiovascular disease and clinical CVD outcomes	**Y1**		**Y2**		**Y3**		**Y4**		**Y5**	
Monthly meetings with Dr. Sokos										
Advanced Anatomy, Physiology, and Pathophysiology course (NSG 744A)										
Cardiology for Nursing course (NSG 435)										
Echocardiography technician certificate training program sessions										
Individualized cardiac imaging training and observation (Dr. Sokos)										
Participation on CARDIA Endpoints Surveillance and Adjudication Subcommittee (Dr. Gabriel)										
10-day Seminar on Epidemiology & Prevention of CVD & Stroke										
Training Goal 3 - Develop skills in grant writing, research management, and leadership	**Y1**		**Y2**		**Y3**		**Y4**		**Y5**	
Monthly grant development meetings with Dr. Smith										
West Virginia Clinical and Translational Science Institute (WVCTSI) PI Academy										
ICOH cardiovascular disease sub-committee and CARDIA physical activity workgroup meetings										
American Heart Association (AHA) Epidemiology|Lifestyle Conference										
American College of Sports Medicine (ACSM) Annual Meeting										
The International Society of Behavioral Nutrition and Physical Activity Annual Conference										
International Commission for Occupational Health (ICOH) Bi-Annual Congress										
West Virginia Clinical and Translational Science Institute (WVCTSI) R Grant Writing Group (15 weeks)										
R01 drafting and initial submission, planned revision, and resubmission as necessary										

E) SPECIFIC AIMS

Cardiovascular disease (CVD) is a major public health problem affecting nearly half of US adults. To reduce CVD risk, the Department of Health and Human Services (DHHS) recommends that adults engage in ≥150 mins/week of moderate-intensity aerobic physical activity (PA). These general recommendations are based on extensive evidence of cardiovascular and other health benefits from leisure-time PA (LTPA) and assume that PA in any context (i.e., domain) is cardioprotective. However, they do not consider recent evidence suggesting that occupational PA (OPA) may not have the same health-enhancing effects as LTPA – and may even have adverse impacts. These differential effects, called the '*PA health paradox*', have been increasingly reported in the past decade, mostly in men and European cohorts, and recently in several secondary data analyses I have led within large US cohorts. If the PA health paradox is true, workers with physically active jobs who primarily accumulate their PA at work (and have less time and capacity for additional LTPA) may be disadvantaged. These workers, who are more likely to belong to low socioeconomic status and racial or ethnic minority groups, may be meeting the PA Guidelines criteria but unfortunately not reaping the expected reductions in mortality and cardiovascular risk. This likely contributes to disproportionately high levels of CVD in these groups.

Despite the important public health implications, the existence and nature of the PA health paradox remains controversial, especially in the US, due to mixed findings and methodological weaknesses. Specific gaps in current research are: (1) poor OPA measurement with highly variable, unvalidated self-report instruments or gross categorization based on job type or characteristics, and (2) use of imprecise outcomes (e.g., all-cause mortality) that lack direct investigation of underlying biological mechanisms. As a result, the paradox remains poorly understood, delaying robust evaluation for possible updates to PA guidelines and workplace policies to promote health. This application will move the field forward by addressing the primary limitations of the current literature by (1) evaluating the validity of common self-report OPA measures and identifying the optimal self-report measure(s) to use in future large-scale research and (2) leveraging existing NHLBI cohort data to interrogate proposed but untested explanatory biological mechanisms of the PA health paradox. My aims are:

Aim 1: Identify the optimal self-report measure(s) of work-related activity, job characteristics, and activity recall to approximate device-based measures of OPA

OPA research is limited by poor measurement. While device-based OPA measurement with concurrent time-use diaries to identify work periods is considered best-practice, this method is rarely used in large studies due to participant burden and cost. Studies often rely on self-reported OPA data (e.g., job characteristics and typical work-related activity) using a variety of questionnaires. Typically, these instruments are not purpose-built or validated against devices. Previous day PA recalls are also available and may be more valid but are infrequently used in OPA research. Overall, OPA measurements are inconsistent, which could explain heterogeneous associations with health outcomes. To fill this methodological gap, I will conduct an original, observational study to concurrently measure OPA using state-of-the-art devices, commonly used self-report questionnaires, and previous day PA recall in n=50 workers. A primary deliverable of this project will be to identify the optimal self-report measure(s) to approximate device-based OPA metrics. This validation will inform rigorous OPA exposure estimation and OPA-risk assessments in future large-scale studies. I hypothesize that:

H1: OPA levels from device-based measurement can be accurately estimated using self-reported work-related activity questions and previous day activity recalls.

Aim 2: Estimate longitudinal associations of OPA with changes in mechanistically-relevant CVD risk and clinical outcomes within the CARDIA and REGARDS cohorts

Small mechanistic studies, including my own, suggest that chronic cardiovascular strain from OPA may drive adverse changes, but little research has linked chronic OPA exposure to mechanism-specific cardiovascular endpoints, which would strengthen causal inference. The Coronary Artery Risk Development in Young Adults (CARDIA, n~1531 working adults in my proposed analyses) study provides rare, longitudinal self-reported OPA measures across 35 years through mid-life, along with resting blood pressure (BP), heart failure (HF) biomarkers and HF stages. The REasons for Geographic And Racial Differences in Stroke (REGARDS, n=1866 working adults) study includes adults aged ≥45 years with device-based OPA and high-quality measures of resting BP, inflammation, and adjudicated CVD events and mortality. I hypothesize that:

H1: High lifetime OPA will be associated with higher resting BP, HF biomarkers, and HF stage in CARDIA

H2: High OPA will be associated with CVD risk factors (i.e., higher BP and inflammation) and higher incidence of CVD outcomes (i.e., CVD events and mortality) in REGARDS

Impact: This project addresses major gaps in understanding OPA and cardiovascular risk by improving OPA measurement methods and testing biological mechanisms in two diverse US cohorts. The resulting empirical evidence will inform updates to PA guidelines and workplace health policy. The project will also generate tools and training, positioning me to independently lead future R01-level studies in occupational health.

F) RESEARCH STRATEGY

F.1) Significance: Cardiovascular disease (CVD) remains the leading cause of death in the US, despite ongoing attention to prevention and treatment.[18,19] Physical activity (PA) is widely accepted and recommended by the Department of Health and Human Services (DHHS) to prevent CVD.[20–23] Importantly, the PA Guidelines are agnostic to the domain in which the PA occurs.[21] Both leisure-time PA (LTPA) and occupational PA (OPA) count equally towards the recommended ≥150 mins/week.[20–23] However, these recommendations are at odds with recent evidence suggesting a '*PA health paradox'* that posits OPA is paradoxically harmful to cardiovascular health.[24–27] As such, the recent DHHS '*Call to Action*' identified OPA as a key topic area for research and review during the upcoming 2028 PA Guidelines revision.[28]

I have demonstrated that workers in physically demanding jobs often meet DHHS recommended PA levels via OPA alone.[1,12] However, if the PA health paradox is true, those workers, who are often in low socioeconomic and racial or ethnic minority populations already at higher CVD risk, would not experience the expected health benefits from their OPA. As such, studies are needed to investigate OPA as a novel CVD risk factor in workers, with a focus on CVD risk development across the lifespan and potential interventions.[28]

Two primary limitations of the PA health paradox research preclude development of interventions and guidelines supporting cardiovascular health in workers with high OPA. First, OPA is usually measured with crude job classifications or unvalidated and inconsistent self-report instruments.[29] Such self-report measures are susceptible to recall bias and measurement error.[30,31] Previous day activity recalls may have less recall bias and error but are rarely used in OPA-health research.[16] While device-based OPA monitoring is valid and possible[1,2], these methods are not often prioritized, especially in large studies, due to participant and resource burden. Thus, clarifying if self-report and previous day recalls can estimate device-based OPA measurement is critical. My proposal will fill this methodological gap by evaluating the validity of common self-report OPA measures and identifying the optimal self-report measure(s) to use in future large-scale OPA research.

The second major limitation is a lack of rigorous research examining relevant outcomes to interrogate the proposed underlying biological mechanisms. LTPA is generally of short duration and results in lower 24-hour cardiovascular strain via acute post-exercise hypotensive responses and improved cardiorespiratory fitness with repeated exposure. In contrast, it is hypothesized that the sustained cardiovascular strain from high-volume but low intensity OPA does not improve fitness and may catalyze inflammation, vascular damage, and eventual CVD (Figure 1).[24] However, empirical research testing these hypotheses is scarce, often in middle-aged male workers only, and is limited by variable, mostly self-report OPA assessment at a single time point. Available studies lack the precision needed to inform intervention development. My proposal directly addresses these limitations by conducting new analyses in two large, NHLBI-funded, US cohorts with age- and sex-diverse participants, rare state-of-the-art device-based or repeated OPA measurements, and clinically meaningful, mechanistically-relevant outcomes.

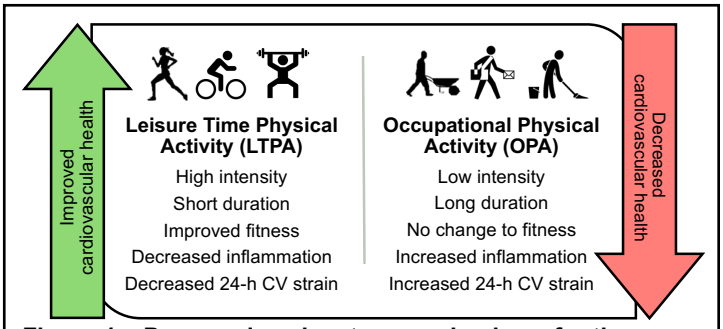


Figure 1 – Proposed explanatory mechanisms for the adverse OPA-CVD associations[24]

This proposed project will identify the optimal self-report tool(s) to measure OPA in future large-scale OPA studies and generate at least four high-impact publications sharing the methodological findings and interrogating the cardiovascular health impacts of OPA on high-quality, mechanism-driven outcomes. The findings will directly inform my long-term research goals to add OPA exposure assessment with longitudinal follow-up of clinically meaningful cardiovascular outcomes in a future cohort (R01 application). This project will directly support my path to research independence as well as inform future workplace policies, intervention development, and the separate consideration of OPA in the 2028 revision of the US PA guidelines.

F.2) Innovation: This project introduces methodological and conceptual advances to progress OPA research and challenge current assumptions in cardiovascular health. The project includes the following innovations:

- Examining OPA as a novel CVD risk factor challenges current thinking, practice, and PA Guidelines.
- Applying and validating best-practice measurement methods from traditional PA epidemiology for OPA research by 1) isolating work periods within device data using time-use diaries, and 2) testing previous day PA recall as a novel assessment tool for OPA exposure.
- This will be the first study to examine OPA with heart failure and subclinical biomarkers which directly align with hypothesized explanatory mechanisms of the PA health paradox.

F.3) Background

I. CVD in the US workforce: CVD imposes a disproportionate burden on low wage workers who are pivotal to US infrastructure. Despite technological advancements, ~34 million workers (21.7%) remain employed in manual labor industries, such as construction, maintenance, and material moving.[32] The prevalence is notably higher in Hispanic or Latino (33.3%) and Black or African American (23.4%) sub-groups (Figure 2).[32]

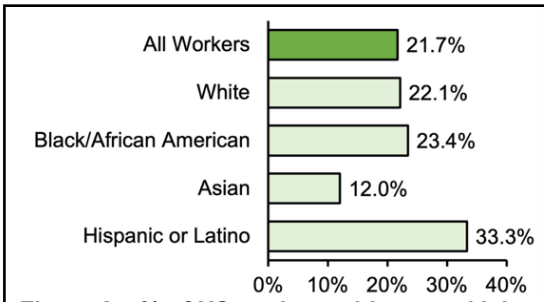


Figure 2 – % of US workers with manual labor jobs by race & ethnicity sub-groups[32]

Health is unacceptably poor in workers with physically demanding jobs. This population faces a 40% higher prevalence of coronary heart disease and stroke compared to those in non-physical labor jobs.[33] Furthermore, each worker with CVD loses an average of seven productive life years, burdening families, employers, and healthcare systems.[34] Exacerbating the problem, workplace health promotion interventions have primarily catered to higher socio-economic status workers, leaving workers with physically demanding jobs underserved.[35–37] If unaddressed, data suggest continued escalation of CVD in these workers, intensifying health differences and costs.[34]

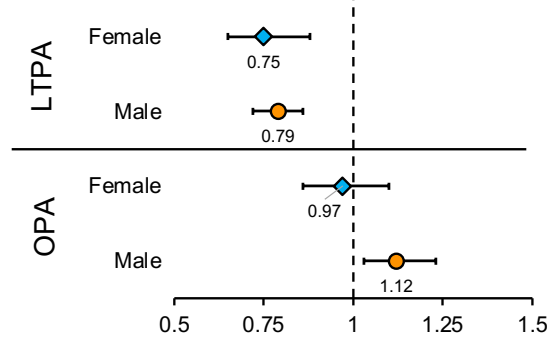


Figure 3 - Meta-analytic hazard ratios for all-cause mortality with high vs. low OPA & LTPA[26]

II. A gap in CVD prevention, the '*PA Health Paradox*': My research addresses a growing concern by focusing on the distinctive health needs of workers with physically demanding jobs to inform effective CVD prevention strategies. I focus on a novel but potentially high-impact CVD risk factor: OPA. The current DHHS PA Guidelines recommend ≥150 minutes/week of moderate-to-vigorous intensity aerobic PA to promote cardiovascular health.[21] Important to workers, OPA counts equally towards this goal.[21] I demonstrated that many workers in highly active industries meet the PA target via OPA alone[1,12] and are less likely to be active during leisure.[38] Yet challenging the assumption that all PA is beneficial in the DHHS PA Guidelines, emerging evidence suggests that OPA, particularly when high-volume and repetitive, may not confer similar health benefits achieved through LTPA and may even exacerbate CVD risk.[27,29,39] For example, OPA is not protective against CVD mortality and may even increase all-cause mortality risk in males (Figure 3).[26,27,29,39–41] Although cross-sectional or examining limited outcomes, my preliminary studies demonstrate similar adverse cardiovascular impacts of OPA even after robust adjustment for confounding.[11,14,15] These paradoxical associations, i.e., the "PA health paradox", hypothesize that workers with active jobs may meet PA goals through OPA but not receive anticipated cardiovascular health benefits.

Development of targeted interventions and updated PA Guidelines is hampered by two primary limitations: (1) Inconsistent and unvalidated OPA measurement and (2) lack of rigorous research examining the proposed underlying biological mechanisms [11,27,29,39,40] By addressing these gaps, I seek to identify optimal self-report OPA measurement methods for large-scale research and empirically examine the novel and complex associations between OPA and CVD, preparing for future studies, guidelines, and targeted interventions.

Table 3 - OPA constructs measured in 2024 OPA-Mortality meta-analysis[26]

OPA Construct Measured	n
Energy expenditure (e.g., METs, kcal)	4
Lifting and/or carrying	10
Specific work tasks (e.g., digging)	4
Posture (e.g., sitting, standing)	18
Ambulation (e.g., walking, stairs)	17
Perceived physical demand	7
Job classification-based proxy	2

n = # of studies. N=22 studies in the review and each study measured multiple OPA constructs.

III. Limitation 1 – Poor OPA measurement: OPA is typically poorly categorized in large cohorts using highly variable, and often unvalidated, self-report instruments that measure a wide variety of OPA constructs such as energy expenditure, lifting, posture, ambulation, and job classification (Table 3).[26,29,30] Not only do self-report measures introduce recall bias and measurement error,[30,31] the variety of methods used can make harmonization difficult or inappropriate, precluding reliable conclusions.[26,29] Previous day activity recalls such as the Activities Completed over Time in 24 hours (ACT24) ask participants to recall their previous day's PA by domain (e.g., work, leisure).[16,17] While not widely used in OPA-health research, previous day recalls may limit bias from traditional questionnaires by lowering reliance on long-term memory and the cognitive complexity of summarizing typical exposure with variable OPA.[16] Lastly, state-of-the-art device-based OPA monitoring is possible and highly valid, yet has not been commonly utilized, especially in large studies, due to participant and resource burden.[1,2,26] Given the variety of methods used, there is a clear need for improved and

standardized self-report OPA measurement methods with calibration to best-practice device-based metrics.[31,42]

IV. Limitation 2 – Unexplored explanatory mechanisms: Secondly, research elucidating the underlying explanatory mechanisms differentiating the effects of OPA from LTPA is critical to support biological plausibility but remains largely unexplored. OPA is often of low-to-moderate intensity with long duration, which may lead to sustained cardiovascular strain, triggering chronic inflammation and cardiovascular damage, and increasing CVD risk.[15,24,40] Although preliminary studies, including my research, indicate that high OPA is associated with acute cardiovascular strain, diminished fitness, and impaired cardiac function, existing research is based on limited data from small homogeneous samples.[2,14,15,43,44] Most research in large samples has relied on broad mortality outcomes not clearly linked to biological mechanisms or has utilized self-reported OPA at one time-point as a proxy for long-term OPA exposure.[27,29] Reflective of the limited mechanistic research, the causal foundation for the paradox is lacking. As such, studies directly examining associations between rigorously measured OPA and biologically plausible CVD outcomes are urgently needed.

V. Need for rigorous, empirical OPA research: These two limitations make the presence and nature of the PA health paradox unclear, restricting scientific advances to reduce the possible risks from OPA. Thus, there is an unequivocal need for rigorous research to improve available OPA measurement methods and subsequently test the proposed explanatory mechanisms with biologically-relevant outcomes. My project will do this by (1) evaluating the validity of common self-report and previous day OPA measures against state-of-the-art device-based metrics to identify the optimal self-report measure(s) for use in future large-scale research and (2) interrogating proposed but untested biological mechanisms of the paradox via relevant cardiovascular endpoints in existing NHLBI-funded cohorts. This research will strengthen or oppose the PA health paradox hypothesis, informing future occupational health policies, interventions, and CVD prevention strategies.[45]

F.4) Preliminary Studies

I. Descriptive epidemiology of OPA and health: I conducted preliminary studies of the PA health paradox in several populations. In the CARDIA cohort, I demonstrated that PA patterns are partly determined by one's occupation.[12] In National Health Interview Survey data, I observed that those with high self-reported OPA are about twice as likely to have CVD compared to not reporting OPA, stratified by sex and smoking status.[11] However, these studies relied on cross-sectional designs and limited OPA exposure assessments.

II. Pilot studies of mechanistically relevant CVD outcomes: Next, I examined the proposed biological mechanisms in a pilot study (n=19) with short-term observation of high-quality, device-based OPA and cardiovascular responses. The study was the first to rigorously demonstrate that 1) workers with high OPA meet or exceed DHHS recommended PA levels during worktime alone[1] and 2) within subjects, workdays with high OPA had elevated cardiovascular strain (i.e., 24-hour heart rate [HR] and BP, worse HR variability [HRV]) compared to non-workdays without OPA.[2] Then, I examined CVD outcomes in the pathway of the proposed biological mechanisms in the CARDIA cohort. In contrast to LTPA which improved cardiorespiratory fitness, high OPA resulted in lower fitness over 13 years[14] and lower ejection fraction over 25 years (Figure 4).[15]

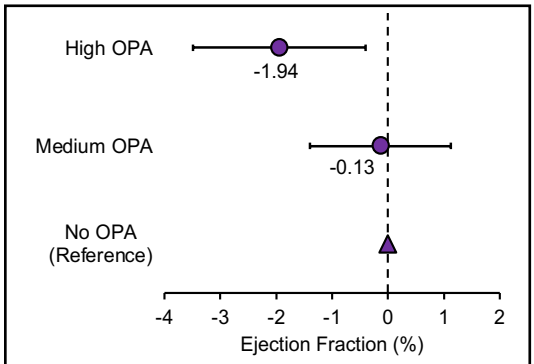


Figure 4 – Preliminary CARDIA study of ejection fraction across self-reported OPA[15]

These initial studies suggest that high OPA might cause acute cardiovascular strain, potentially leading to reduced fitness and CVD, and aligning with the PA health paradox hypothesis. However, these data have notable limitations: either small samples with short-term, high-quality OPA measures or large samples with longer follow-up but imprecise, self-reported OPA. To inform public health action, robust studies with precise OPA assessment, large age-diverse samples, and relevant clinical CVD outcomes are essential.

G) APPROACH

G.1) Aim 1: Identify the optimal self-report measure(s) of work-related activity, job characteristics, and activity recall to approximate device-based measures of OPA

I. Aim 1 Design and Measures: I will conduct an observational study in n=50 participants with equal distribution across no, low, and high OPA jobs at screening based on occupation classifications from previous OPA descriptive research.[12,38,46] Participants will (1) complete a series of commonly used questionnaires about their typical OPA and work characteristics, (2) the ACT24 measurement protocol, including previous day PA recall on two workdays and one non-workday, and (3) a best-practice protocol for measuring OPA metrics using a thigh-worn accelerometer (activPAL3) with a concurrent time-use diary of work times. I hypothesize that OPA levels from best-practice device-based OPA assessment can be accurately estimated using self-

reported instruments and/or the ACT24.

Findings will provide invaluable information for evaluating the rigor of published/future research using these self-report methods as well as selecting the optimal self-report instrument(s) for future research on OPA and heath if device-based measures are not feasible.

Table 4 – OPA constructs in Aim 1 questionnaires

	Energy expenditure	Lifting/carrying	Specific work tasks	Posture	Ambulation	Perceived demand	Job classification
Census			X				X
OPAQ				X	X	X	
BRFSS				X		X	
IPAQ-L	X	X		X	X	X	
TOQ	X			X	X		
ACT24			X	X	X		

Self-reported OPA measurement: I will include a suite of common self-reported questionnaires to capture a variety of OPA constructs (e.g., energy expenditure, work tasks, and posture) and work-related demographics (e.g., work hours and job classifications) which have been used in previous research of OPA-cardiovascular health associations. I will include the US Census standard questions collecting information about industry, occupation, common tasks, and tenure[47] as well as OPA-specific questions from four commonly used questionnaires[31,42] covering a variety of OPA constructs (Table 4): (1) Behavioral Risk Factor Surveillance System (BRFSS) OPA questions[48] (2) International Physical Activity Questionnaire – Long Version (IPAQ-L)[49], (3) Tecumseh Occupational Activity Questionnaire (TOQ)[50], (4) Occupational Physical Activity Questionnaire (OPAQ)[51].

Previous day activity recall: Participants will be asked to complete the Activities Completed over Time in 24 hours (ACT24) Version 2 for three days (one non-workday and two workdays).[16,17] The ACT24 is an online module where participants select from 170+ activities (12 specific to work) in 14 categories to recall all activities completed across a 24-hour period on the previous day.[16,17] All activities are mapped to the Bureau of Labor Statistics' American Time Use Survey and Compendium of PA categories and reported by domain (e.g., work, leisure).[52–54] Three measurement days will allow for stable estimates of long-term behavior.[55]

Device-based OPA measurement: I will employ rigorous, state-of-the-art, device-based OPA measurement via a thigh-worn activPAL3 micro which is highly valid for measurement of posture and ambulation in general populations.[56] The best-practice protocol includes 7-days of 24-hour wear with a concurrent time-use diary of work, sleep, and wear times to precisely isolate work periods within the data. Data will be processed with a custom Python program to integrate accelerometer data with diary-reported times. The data will then be summarized as daily work time spent in sedentary behavior (SB), standing, light-intensity PA (LPA), and moderate-to-vigorous intensity PA (MVPA) as well as total steps during work.

II. Aim 1 Analytic Approach: To determine the most valid self-reported OPA measures, I will compare survey- and recall-based estimates of work activity times and durations to the criterion outcomes of device-based OPA metrics (i.e., SB, standing, LPA, MVPA, steps). I will perform a series of correlations and linear regression models using individual self-reported and previous day recall OPA measures to predict device-based metrics described above. Model performance will be evaluated using the Akaike and Bayesian information criterion (AIC and BIC) and classification accuracy will be examined using adjusted Rand index and confusion matrices. I will also compare the self-report and device-based metrics using the Bland-Altman methods.[57] Secondary analyses will examine different combinations of self-report and/or recall metrics for accurately predicting device-based estimates with a goal to identify whether a suite of questions provides the best model fit. Lastly, I will explore whether predictive performance across all models varies by participant characteristics (e.g., sex, BMI, occupation). Results will identify the best self-report OPA measurement methods for use in future studies.

III. Aim 1 Power Analysis: Though our validation study is distinct because it will isolate accelerometry time to work periods, a previous review evaluated the validity of self-reported OPA against accelerometer-measured physical activity across the entire day.[31] Reported correlations ranged from r = -0.20 to 0.77, with higher correlations observed for specific constructs such as sitting and light activity.[31] Because our validation will be a more direct comparison between reported and device-based OPA metrics during isolated work periods, I anticipate correlations of at least r = 0.40. An a priori power analysis using a bivariate normal correlation test indicates that a sample of n=46 provides 80% power to detect a r = 0.40 (two-sided $\alpha = 0.05$). To account for possible incomplete data, I will recruit a sample of n=50 which is consistent with similar calibration studies.[31,42]

G.2) Aim 2: Estimate longitudinal associations of OPA with changes in mechanistically-relevant CVD risk and clinical outcomes within the CARDIA and REGARDS cohorts

I. Aim 2 Summary: To directly fill the research gap in testing the proposed biological mechanisms for the PA health paradox, Aim 2 will utilize two, large, NHLBI-funded cohorts to rigorously examine longitudinal associations of OPA with changes in subclinical CVD risk and clinical CVD outcomes. Study 1 will examine the The Coronary Artery Risk Development in Young Adults (CARDIA) cohort and study 2 will examine the REasons for Geographic And Racial Differences in Stroke (REGARDS) cohort. Findings from this aim will

improve causal inference by evaluating hypothesized mechanistic pathways through which OPA leads to CVD.

II. Study 1 - Design and Measures (CARDIA cohort): The CARDIA cohort (analytic sample: n~1531 working adults) is a prospective, multicenter cohort investigating the development and determinants of CVD risk factors in young adults.[58,59] The study enrolled 5,115 black and white men and women aged 18-30 years in 1985-86 from four US cities. To date, participants have been followed for >35 years, with in-person exams at years 0, 2, 5, 7, 10, 15, 20, 25, 30, and 35. The Activity and HF Ancillary (R01HL149796, PI: Dr. Gabriel, co-mentor) has analyzed stored blood at years 20, 30, and 35 for HF biomarkers. I will examine trajectories of self-reported OPA from years 20 to 35 with changes in HF biomarkers (NT-proBNP and hscTnT) in the same period. I will also examine OPA trajectories with changes in resting BP and HF stage from years 5 to 30. I hypothesize that high OPA will be associated with adverse changes in cardiovascular risk. All study measures are in Table 5.

OPA measurement: Using our previous methods in CARDIA[14,15], I will utilize the CARDIA PA questionnaire collected at every exam to determine OPA levels. OPA exposure will be determined as the number of participant-reported months/yr with "vigorous job activities such as lifting, carrying, or digging" for ≥five hours/week. The number of months at each exam will be used to calculate OPA trajectories across the relevant study period. OPA trajectory group will then be used as the predictor in statistical models.

Table 5 - Measurements used in proposed Aim 2 studies

		Study 1 CARDIA Cohort		Study 2 REGARDS Cohort	
		Baseline	Follow-up	Baseline	Follow-up
OPA Exposure	Accelerometry			✓	
	Time-use Diary			✓	
	Self-reported OPA	✓	✓		
CVD Risk Factors	Resting BP	✓	✓	✓	✓
	C-Reactive Protein			✓	✓
	HF Biomarkers	✓	✓		
CVD Outcomes	HF Stage(s)	✓	✓		
	CVD Events				✓
	Mortality				✓

BP measurement: Resting BP was measured three times during each exam on the right arm at 1-minute intervals after a 5-minute rest. The average of the second and third measurements will be used for analyses.

HF biomarkers and stage measurement: HF biomarkers (NT-proBNP and hscTnT) were measured at CARDIA exam years 20, 30, and 35 and will be treated continuously. Additionally, I will utilize the AHA HF stage calculated from the year 5 and 30 exam echocardiography. HF stages are defined using the 2013 AHA/American College of Cardiology guidelines[60]: no risk (stage 0), at risk (stage A), presence of cardiac structural/functional abnormalities (stage B), or symptomatic/end stage (stages C/D).

Covariate measurement: Age, sex, race, center, education, income, smoking status, alcohol consumption, LTPA, antihypertension medication use, and diabetes status are routinely self-reported by participants at each exam. Height and weight were measured during each in-person exam.

III. Study 1 - Analytic Approach: HF biomarkers will be examined using multivariate generalized linear models testing the association of 15-year OPA trajectories with biomarkers at follow-up (year 35) after adjustment for year 20 baseline values. Exploratory analyses will also examine OPA trajectories including all exposures assessments and HF biomarker changes (years 20-35). Changes in HF stage will be tested using logistic regression models examining the odds of developing stage A/B or stage C/D at year 30 (when echocardiography was assessed) within each OPA trajectory group.[61] Lastly, I will examine the possibility of confounding and effect modification by socio-demographic (i.e., sex, age, and race) and behavioral factors (i.e., LTPA, smoking status, and alcohol intake) with covariate adjustment, interaction terms, and stratification.

IV. Study 1 - Sample Size and Power: In CARDIA, n=1935 completed the in-person exam component in year 35 and, based on previous CARDIA analyses[15], we expect n=1531 to be working with unequal distribution across no, medium, and high OPA groups (sample sizes in Table 6). I conducted a priori power analyses (power=0.80, α=0.05) to assess detectable differences in resting systolic BP across OPA groups. The sample-specific standard deviation for systolic BP is based on prior CARDIA studies.[14,15] I estimated omnibus and pairwise effect sizes, applying a Bonferroni correction (α=0.0167) for comparisons between the no and high OPA groups. A 5 mmHg reduction in systolic BP is considered clinically meaningful due to its association with reduced CVD risk.[62] The estimated minimum detectable effect size in CARDIA is 2.31 mmHg which falls below this threshold, even after adjusting for multiple comparisons. Thus, the study is sufficiently powered to detect meaningful group differences if present, including potential differences by sex. Details in Table 6.

The CARDIA HF biomarkers are only available in n=671 (NT-proBNP) and n=663 (hscTnT) at both timepoints. A similar power analysis as above resulted in omnibus minimum detectable differences of 135 pg/mL for NT-proBNP and 3.5 ng/L for hscTnT, respectively. These detectable differences are much smaller than the accepted threshold for a clinically meaningful changes in NT-proBNP (300-400 pg/mL[60,63,64]) but greater than the meaningful change from baseline for hscTnT (>20-50%[65–67] or ~1.2-3.1 ng/L in CARDIA).

V. Study 2 - Design and Measures (REGARDS cohort): The REGARDS cohort (analytic sample: n=1866 working adults) is an ongoing, population-based, longitudinal study of 30,239 black and white US adults, aged ≥45 years, enrolled in 2003-7.[68] REGARDS investigates causes of regional and black-white differences in stroke mortality. Participants completed a telephone interview and in-home clinical exam at enrollment and 10-year follow-up. Participants also completed telephone interviews every 6 months to monitor new medical diagnoses that prompt records retrieval and adjudication for a comprehensive set of health outcomes. I will use device-measured OPA as the primary exposure (described in detail below). I will examine longitudinal associations of OPA at baseline with CVD risk factors (resting BP and inflammation [high-sensitivity C-reactive protein (hsCRP)]) measured at 10-year follow-up. I will also examine longitudinal associations with CVD outcomes (CVD events and mortality) collected during the regular 6-month follow-up interviews after OPA measurement (~14 years follow-up for analyses). I hypothesize that high OPA will be associated with higher resting BP, inflammation, and incidence of CVD events and mortality. All study measures are in Table 5.

OPA measurement: Data collection for the REGARDS PA Accelerometer ancillary study occurred between 2009-13, just prior to the clinical exam. Participants wore an Actical accelerometer on their hip during waking hours for seven days and completed a paper time-use diary of sleep and work times. Ultimately 8096 participants had usable accelerometry data (≥4 days with ≥10 wear hours) and a completed diary.[69,70] I will reprocess the accelerometry data to identify the specific periods of OPA with the diary. As previously done in REGARDS, 60-second epochs will be classified into SB, LPA, and MVPA using standardized cut points.[69,70] Utilizing the unique but unused time-use diary data, I will also re-categorize activity epochs as OPA or LTPA.[1,2]

BP measurement: Resting BP was collected according to a standardized protocol (in duplicate following a 5-minute seated rest period) during baseline and 10-year follow-up exams. Mean BP values will be used.

hsCRP measurement: hsCRP was measured in plasma with a validated high-sensitivity, particle enhanced immunonephelometric assay at baseline and 10-year follow-up using standardized procedures.

CVD outcomes measurement: Outcomes include incident CVD events (myocardial infarction, stroke, and HF hospitalization) and mortality (all-cause and CVD-related). All outcomes were measured during the 6-month interviews starting after baseline, where participants reported hospitalizations. Medical records were adjudicated by at least two physician-investigators.[68,71] Deaths and their causes were reported by participant proxies or detected through Web searches, the Social Security Death Index, and the National Death Index.[68,71]

Covariate measurement: Participants reported sociodemographics, lifestyle factors, and medical history during baseline and 10-year follow-up interviews. Height and weight were measured during in-home exams.[68]

VI. Study 2 - Analytic Approach: The analyses for Aim 2, study 2 will be like the study 1 approach. I will first classify device-measured OPA profiles using latent-class analyses to use as the independent variable in all models.[72] I will examine associations between OPA profile and changes in CVD risk factors (BP and hsCRP) using linear regression and associations with CVD outcomes using Cox models. In both cases, four nested models with covariate adjustments as well as interaction and stratified models will examine effect modification by socio-demographic (i.e., sex, age, and race) and behavioral (i.e., LTPA, smoking, and alcohol) factors.

VII. Study 2 - Sample Size and Power: n=5806 completed baseline, follow-up, and both relevant ancillaries. Of these, n=1866 reported working during the ancillaries and will be analyzed. Using self-reported job physical effort, I anticipate identifying three unequally distributed OPA profiles outlined in Table 6. Under similar conditions used for study 1, I conducted an a-priori power analysis for systolic BP in study 2. The minimum detectable difference in no versus high OPA is expected to be 4.43 mmHg which is below the clinically meaningful threshold of 5 mmHg, suggesting sufficient power for the analyses.

Based on previous REGARDS samples and anticipated increasing event rates, I expect the all-cause mortality, CVD mortality, and CVD event rates to be at least 32%, 13%, and 19%, respectively for our analyses.[71,73] Assuming 80% power and α=0.05, the detectable hazard ratios comparing high to no OPA groups are 1.52, 2.06, and 1.77, respectively. Though power to detect smaller hazard ratios would be preferred, the precision of device-based OPA assessment improves our chances of detecting true associations and we will report *post hoc* power in our publication to acknowledge the possibility of type 2 errors.

Table 6 – Aim 2 samples & power

	Study 1 - CARDIA[b]	Study 2 - REGARDS
Sample size[a]	1531	1866
No OPA[b]	806 (52.7%)	385 (20.6%)
Medium OPA[b]	429 (28.0%)	1247 (66.8%)
High OPA[b]	295 (19.3%)	234 (12.5%)
Baseline age	29.9 (3.6)	64.3 (7.5)
Female	1092 (53%)	988 (53.0%)
White race	1162 (56.3%)	1331 (71.3%)
Baseline BMI	25.7 (5.2)	28.8 (5.8)
Omnibus comparison power calculation		
σ for systolic BP (mmHg)	10.5	16.4
Effect size (Cohen's f)	0.079	0.072
Min detectable diff (mmHg)	0.82	1.18
Post-hoc pairwise power (no vs. high OPA)		
Effect size (Cohen's d)	0.22	0.27
Min detectable diff (mmHg)	2.31	4.43

Footnotes: Values presented as mean(SD) or n(%). (a) estimated sample sizes for employed participants who have all necessary study measures. (b) estimated using previous CARDIA analyses or self-reported current job physical effort in REGARDS. σ = population standard deviation.

G.3) Expected Outputs and Outcomes: Investigating my hypotheses will result at least four peer-reviewed publications and specific recommendations for which self-report OPA measures will best approximate device-based OPA measurement in future large-scale studies (Table 7). I will share the results at national presentations to both researchers and public health practitioners to promote uptake to occupational health practice and policy. The research outputs will advance understanding of the association between OPA and worker health, broadly, and cardiovascular health, specifically. Findings will answer the recent DHHS call to action for research on OPA and health[28] to inform possible needed revision of existing DHHS PA Guidelines address the health protection and health promotion needs of workers with active jobs. My integrated data-sharing plan will increase the reproducibility of my developed research methods and facilitate further secondary analyses as well as utilization of optimal self-report instruments in larger cohorts. I will also engage in community and stakeholder outreach, including presentations at conferences and working group meetings. With my involvement in the forthcoming revisions of DHHS PA Guidelines, my timely insights have the potential to influence national health recommendations, ensuring they reflect the nuanced impacts of different types of PA. By concentrating on workers who are disproportionately from socioeconomically disadvantaged backgrounds, I will advance policies supporting health equity. Educational outreach will further empower workers and health practitioners to mitigate OPA risks. This foundational research has the potential to yield long-term benefits such as improving the health and productivity of the workforce, reducing CVD rates and mortality, and informing the development of future workplace health standards.

G.4) Scientific Rigor: The first Aim of this project seeks to strengthen the rigor of OPA measurements by using a standardized validation protocol with concurrent device-based, self-reported, and recall measures along with objective model evaluation metrics to identify optimal self-report OPA measures for future studies. Aim 2 leverages two large, NHLBI-funded longitudinal cohorts to examine hypothesis-driven associations of OPA with sub-clinical and clinical CVD outcomes, using validated biomarkers and adjudicated endpoints. Analyses will systematically address confounding, effect modification, and exposure heterogeneity.

G.5) Anticipated Challenges and Pitfalls: For Aim 1, recruitment and compliance may be challenging due to the burden of device wear and time-use diary completion among workers with full-time jobs. To mitigate this, I will implement standardized protocols for device-based PA measurement and diary completion developed by co-mentors Drs. Gibbs and Gabriel and successfully used in previous studies, including user-friendly materials, structured reminders, and optional phone support. For Aim 2, reprocessing raw accelerometry data with manual diary inputs in REGARDS will require substantial person time to verify and align work periods. I have budgeted for adequate graduate student support time for this. I will also use software I developed to automate accelerometry classification and adjudicate diary times based on movement patterns (i.e., sleep/wake cycles and commuting).[2] Lastly, the coordinating across multiple mentors and study teams may introduce logistical delays. To minimize this risk, I will maintain frequent communication with ancillary study PIs, use shared timelines, and proactively adjust plans in consultation with my mentors. Overall, the project is supported by strong infrastructure, existing tools, and experienced mentors to overcome these anticipated challenges.

G.5) Project Timeline: The project activities and deliverables structured across the five-year award period are outlines in Table 7. The timeline reflects logical sequencing of Aim 1 primary data collection and publication by year 3 to provide pilot data and methods justification for an R-level grant application and revision in years 4 and 5. Aim 2 cohort analyses and manuscripts will be completed in years 3-5. This plan and key milestones aligns will with my training goals, directly supporting my progression toward research independence.

G.6) Future Directions: The research aims and training goals proposed in this application will set the stage for my research agenda to comprehensively examine OPA's role in worker cardiovascular health. The pilot data, recommended measurement methodology, and conclusions from carefully crafted secondary analyses (along with my training goals) are key to a future purpose-built ancillary study application within a large cohort to investigate the nuanced cardiovascular health impacts of OPA. I aspire to inform PA guidelines and eventually, develop more effective, tailored workplace health interventions through observational and intervention studies that seek to mitigate the potential risks of OPA.

Table 7 - Research Timeline for Dr. Tyler Quinn

Aim 1 – Identify optimal self-report OPA measures	**Y1**		**Y2**		**Y3**		**Y4**		**Y5**	
Establish and dry run data collection protocol										
Conduct data collection (n=50)										
Data management and statistical analyses										
Write and submit manuscript w/ recommended methods										
Aim 2 – Examine OPA and CVD in large cohorts	**Y1**		**Y2**		**Y3**		**Y4**		**Y5**	
Establish data use agreements and acquire data										
Study 1 Analyses: CARDIA OPA & CVD risk factors										
Study 1 Manuscript: CARDIA OPA & CVD risk factors										
Study 2 Data Prep: Reprocess accelerometer data										
Study 2 Analyses: REGARDS OPA & CVD										
Study 2 Manuscript 1: REGARDS OPA & CVD risk factors										
Study 2 Manuscript 2: REGARDS OPA & CVD outcomes										

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